

Training Configuration and Summary:

Dataset used: rssi_odom_dataset_12apr25_ajust.parquet

model_type: LSTM
group_sec: 80
timesteps_orig: 60
batch_size: 256
min_units: 32
max_units: 256
units_step: 32
dropout_min: 0.2
dropout_max: 0.3
dropout_step: 0.1
activation_functions: ['tanh', 'relu', 'elu']
optimizers: ['adam', 'rmsprop', 'Nadam']
min_layers: 1
max_layers: 2
max_epochs: 150
hyperband_iterations: 3
group_msec: 80
samples_per_sec: 12
forecast_horizon_sec: 8
forecast_steps: 96
timesteps: 60

LSTM Multi-Step Training Report

Report Generated: 2025-05-15 15:54:25

Training Start: 2025-05-15 15-40-24

Training End: 2025-05-15 15-50-26

Training Duration: 10.04 minutes

Predictions Start: 2025-05-15 15-50-26

Predictions End: 2025-05-15 15-52-07

Predictions Duration: 1.68 minutes

Best Hyperparameters:

Number of Layers: 2

Activation: elu

Optimizer: rmsprop

Dropout Rate: 0.20

MAE: 2.1992

MSE: 7.2533

RMSE: 2.6932

R2 Score: 0.5561

Multi-Step Forecast:

Forecast Steps: 96

MAE: 0.0472

MSE: 0.0032

RMSE: 0.0562

R2 Score: 0.6591

Dataset size before down-sample: 85,269 rows
Dataset size after down-sample: 12,829 rows

First 5 timestamps BEFORE:

1744397085760
1744397085770
1744397085789
1744397085801
1744397085817

...

Last 5 timestamps BEFORE:

1744464971366
1744464971377
1744464971390
1744464971402
1744464971414

First 5 timestamps AFTER:

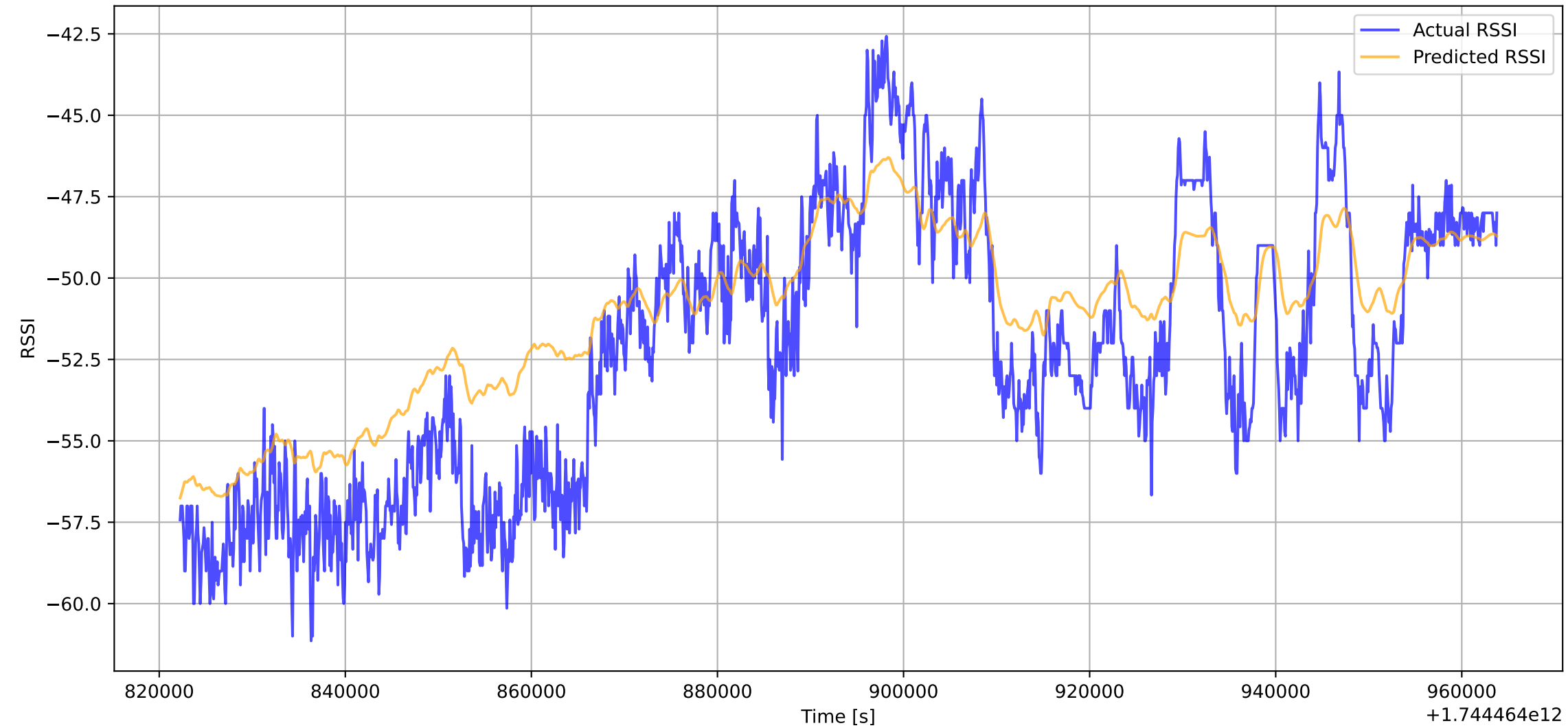
1744397085760
1744397085841
1744397086014
1744397086103
1744397086172

...

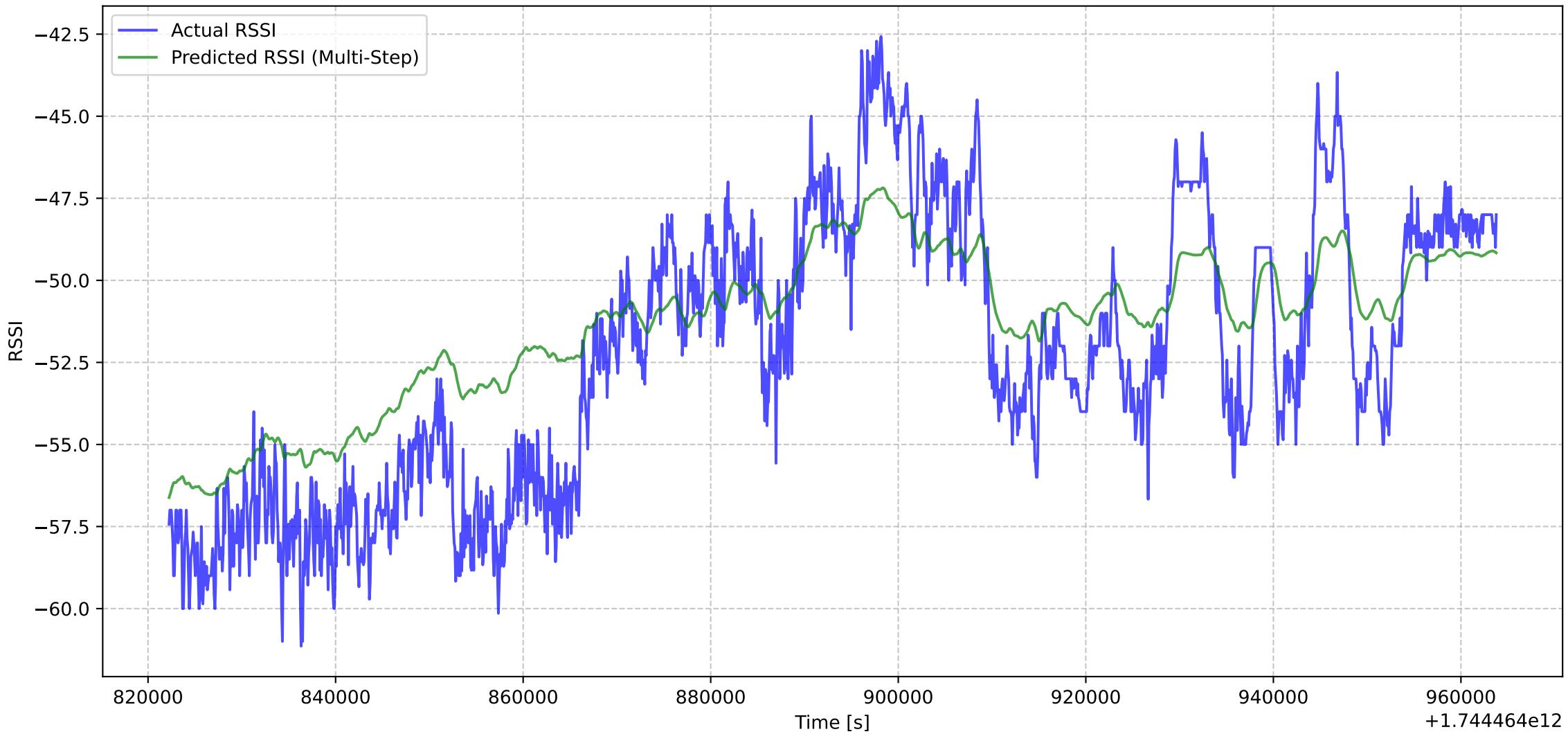
Last 5 timestamps AFTER:

1744464971041
1744464971124
1744464971209
1744464971281
1744464971366

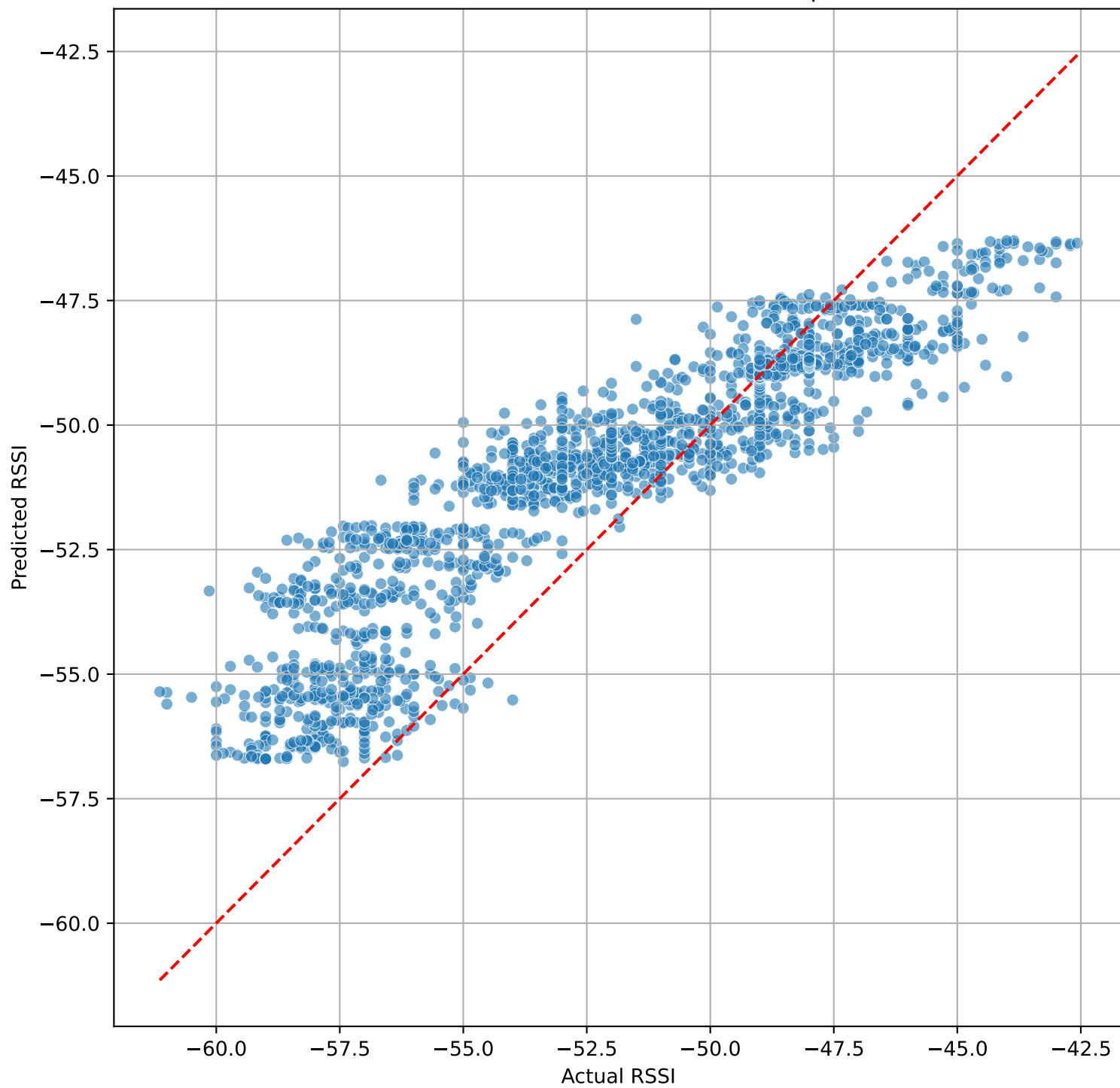
Actual vs Predicted RSSI (First Step)



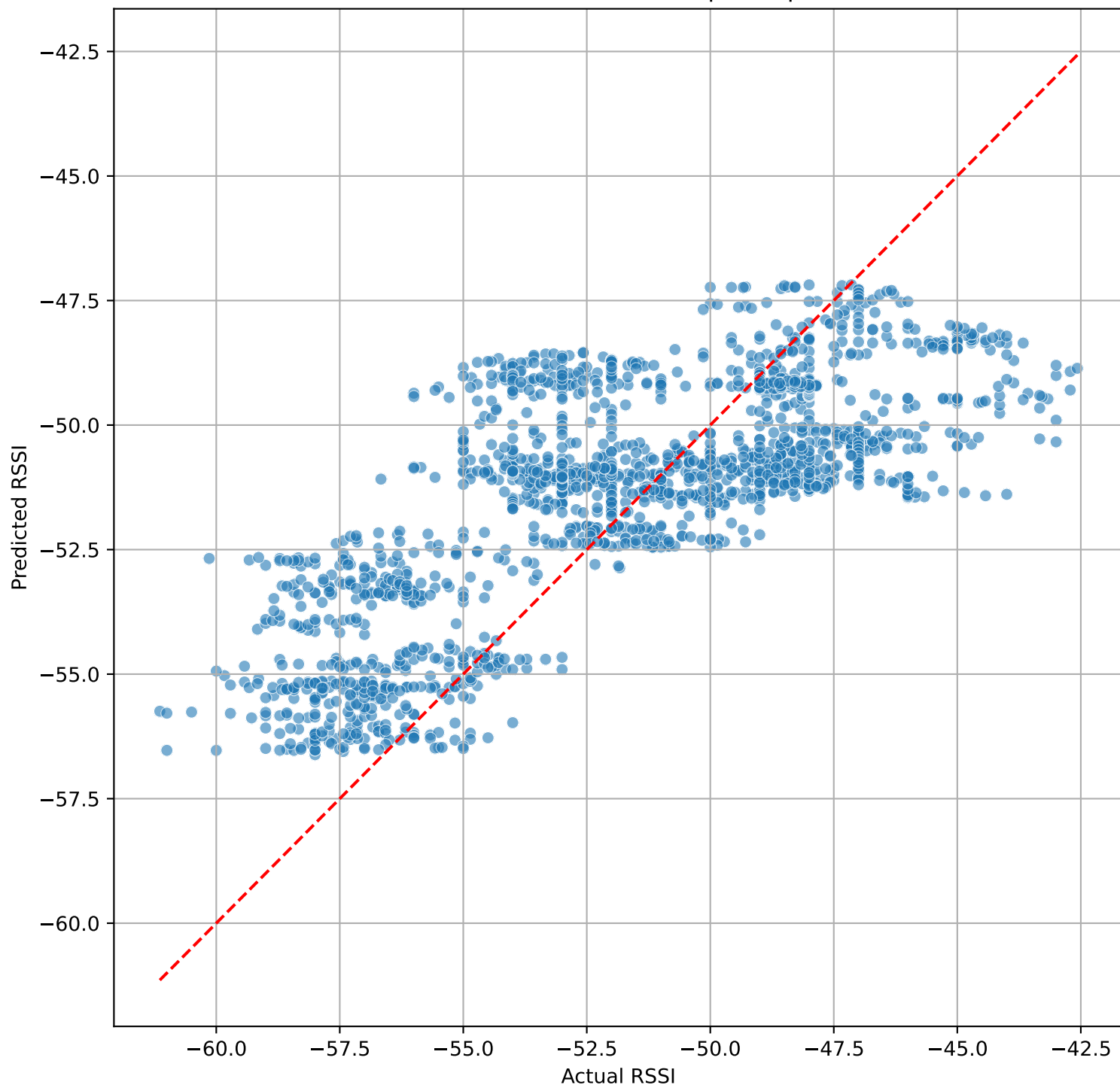
Actual vs Predicted RSSI (Multi-Step (step 96 $\approx$ 8.0 s))



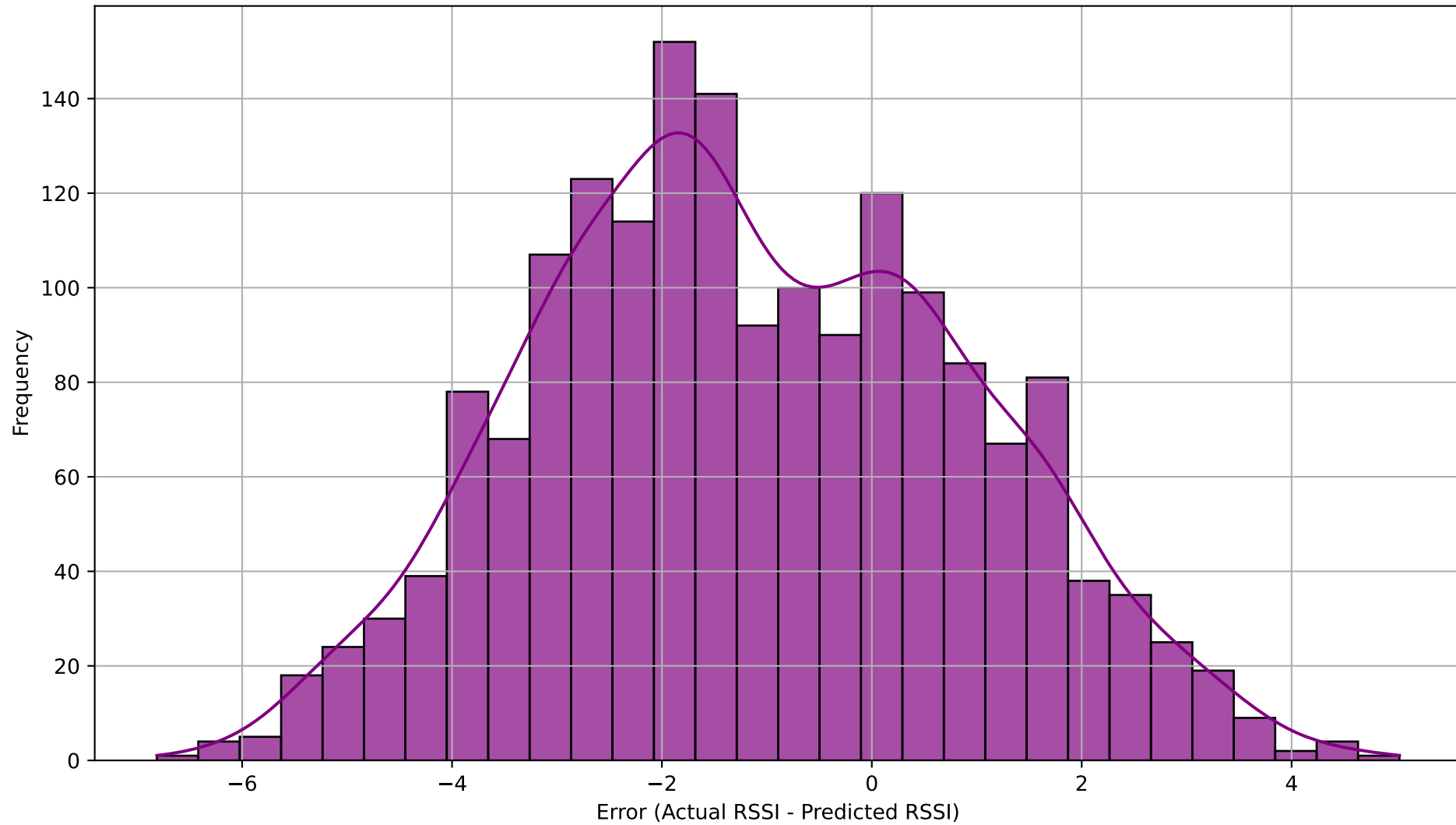
Actual vs Predicted RSSI (First Step)



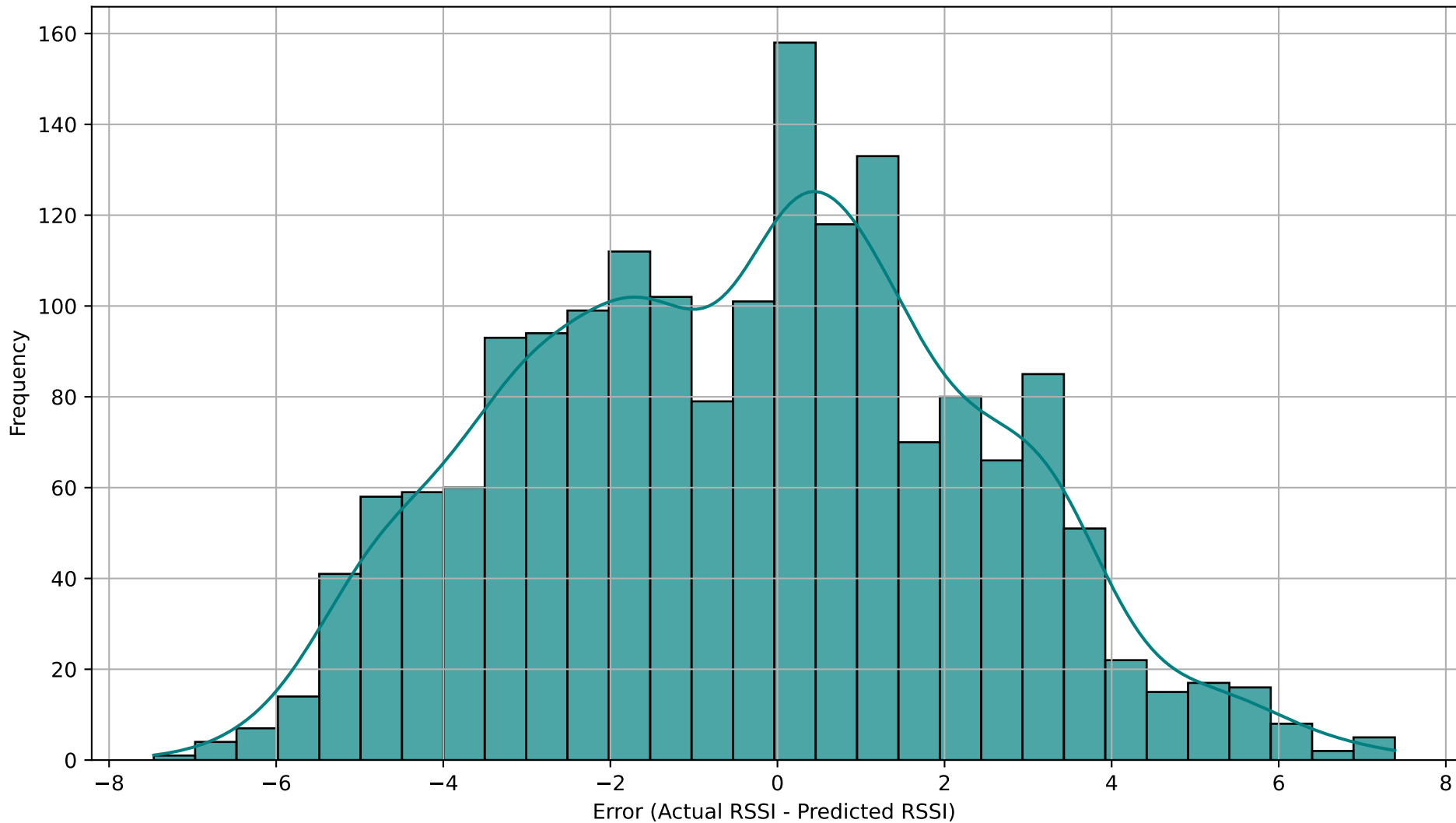
Actual vs Predicted RSSI (Last Step - step 96 $\approx$ 8.0 s)



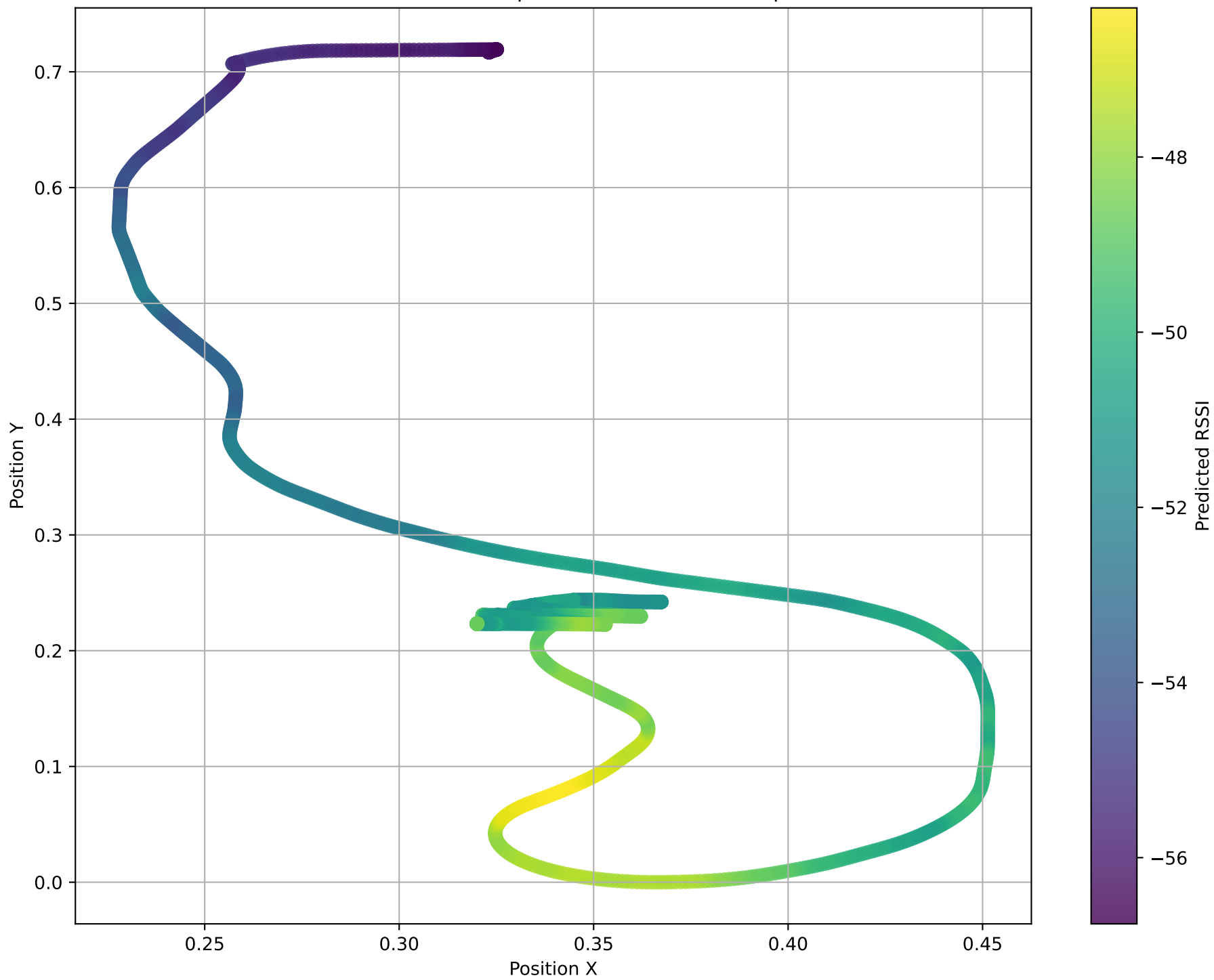
Distribution of Prediction Errors (First Step)



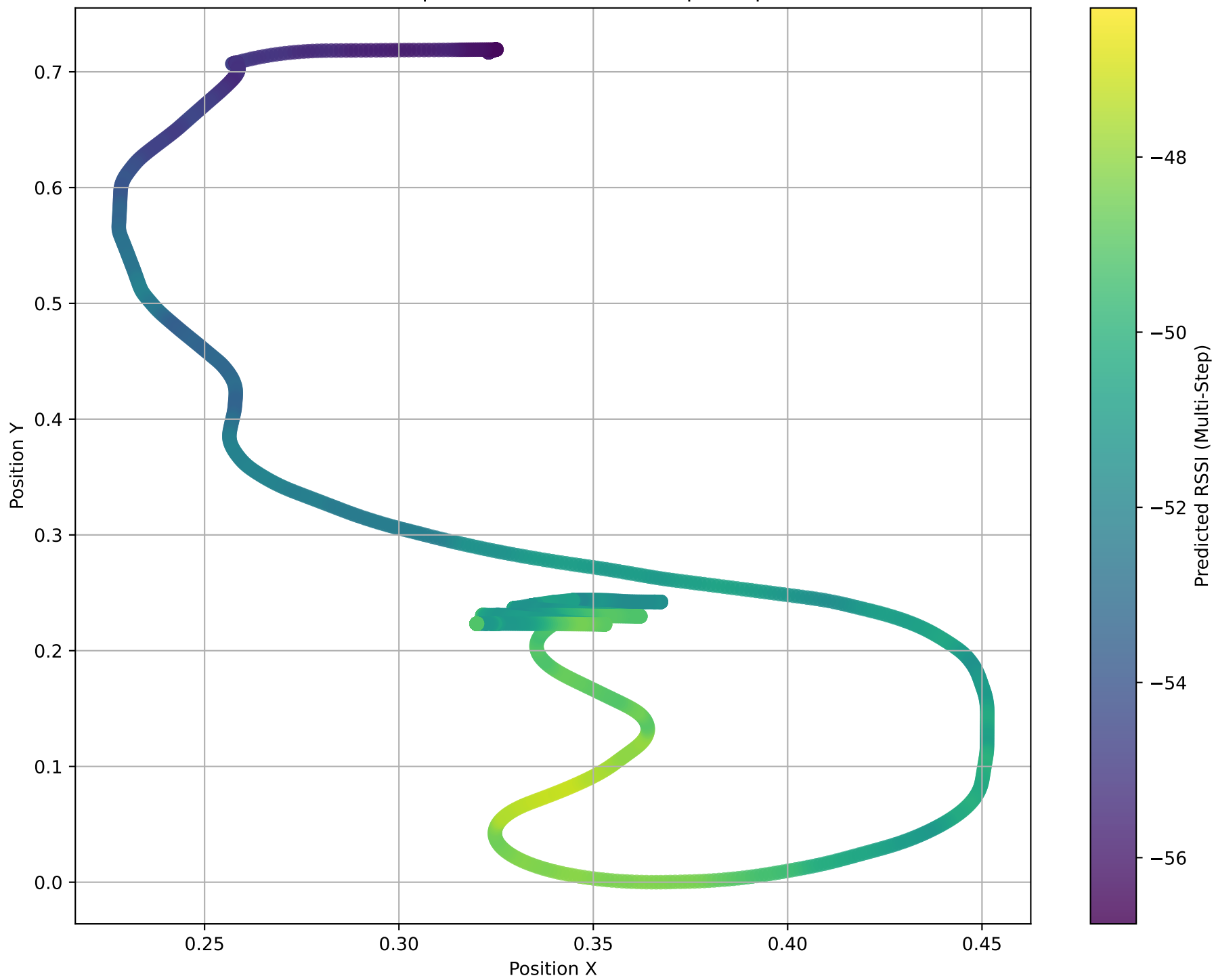
Distribution of Prediction Errors (Last Step - step 96 $\approx$ 8.0 s)



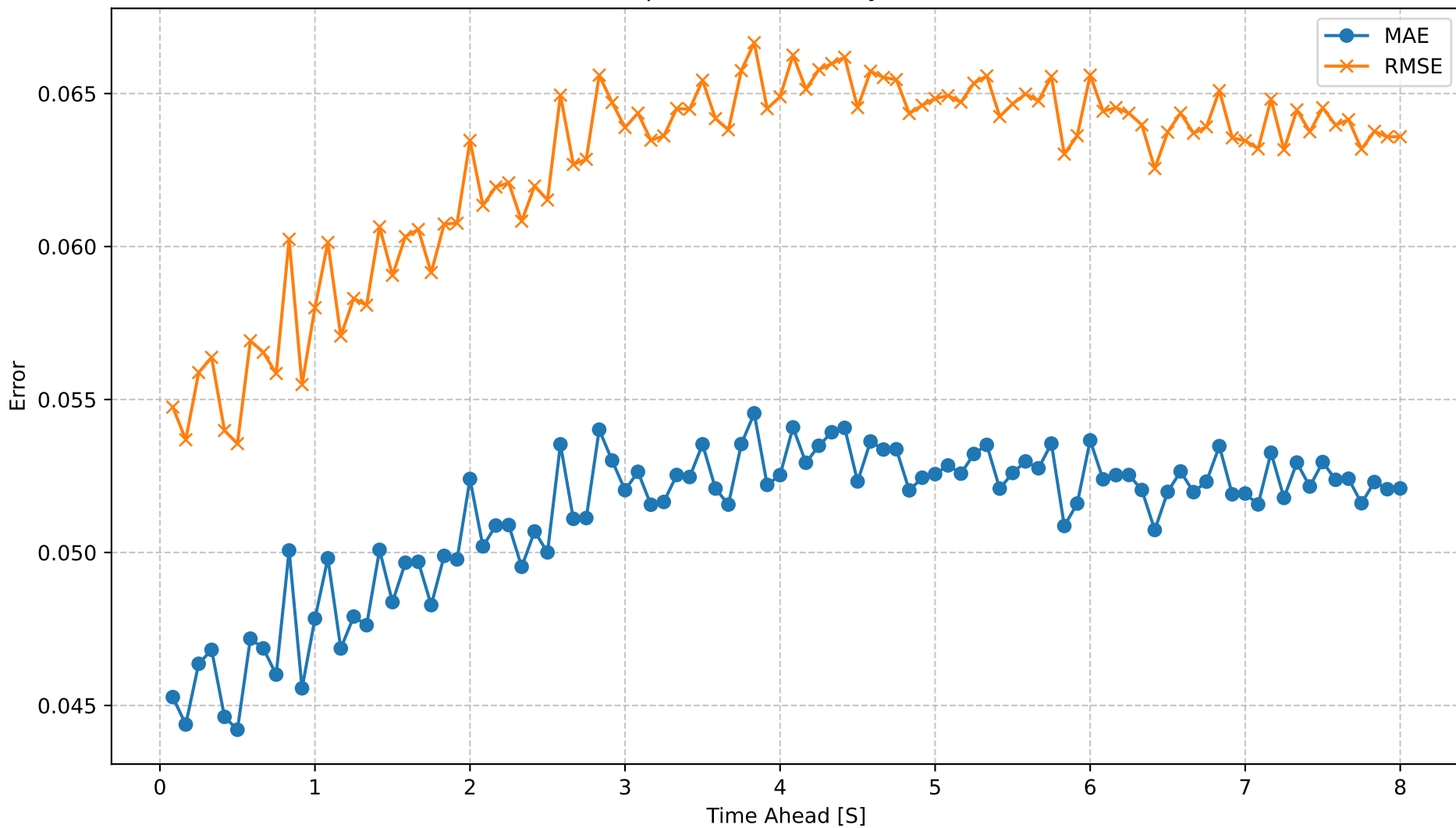
Predicted RSSI in Spatial Context (First Step)



Predicted RSSI in Spatial Context (Multi-Step - step 96 $\approx$ 8.0 s)



Multi-Step Forecast Error by Time Ahead



Resumo Automático

Dataset: rssi_odom_dataset_12apr25_ajust.parquet

Modelo: LSTM

Forecast steps: 96

MAE / RMSE (dB)

- 1º passo : 0.05 / 0.05
- Mediana : 0.05 / 0.06
- 96º passo : 0.05 / 0.06

Crescimento do erro: 15.1% → Estável (erro cresce < 20 %)

3 piores passos (MAE): 46 (0.05), 49 (0.05), 53 (0.05)

Recomendação:

0 modelo é adequado para decisões de hand-over até o horizonte máximo configurado.